

Engineering Mathematics - II

Course Code: BCESH121
Year: I
Program: Civil Engineering

Credit Hours:3
Semester: II
Level: Bachelor

Teaching Schedule Hours/ Week			Examination Scheme				Total Marks	Remarks
			Final		Internal Assessments			
Lecture	Practical	Tutorial	Theory Marks	Practical Marks	Theory Marks	Practical Marks		
3	-	2	60	-	40	-	100	

Course Objective: The basic objective of the course is to provide a sound knowledge of vector calculus, 3-D analytical geometry, Calculus of several variables, infinite series, and ordinary differential equations.

1. Calculus of Several Variables

(4 hours)

- 1.1. Introduction
- 1.2. Partial derivatives, homogeneous function, Euler's theorem for the function of two and three variables, and total differential coefficients
- 1.3. Jacobians and their Properties
- 1.4. Extrema of functions of two and three variables; Lagrange's multiplier

2. Multiple Integrals

(6 hours)

- 2.1. Double integrals in Cartesian and polar form; change of order of integration
- 2.2. Triple integrals in Cartesian, cylindrical and spherical coordinates.
- 2.3. Area and volume by double and triple integrals

3. Analytical Geometry of Three Dimensional

(11 hours)

- 3.1. Plane
- 3.2. Straight line (Co-planer lines, and the shortest distance)
- 3.3. Sphere (Standard equation of sphere)
- 3.4. Right circular cone and right circular cylinder

4. Vector Calculus

(12 hours)

- 4.1. Line integrals, independent of path, conservative and irrotational vector fields, scalar potential
- 4.2. Introduction to Green's theorem and its application
- 4.3. Surface integrals, calculation of flux
- 4.4. Volume integrals, Gauss divergence theorem and its application in evaluation of surface integrals
- 4.5. Introduction to Stake's theorem and its application

5. Infinite Series

(5 hours)

- 5.1. Introduction
- 5.2. Convergence and divergence of a sequence

- 5.3. Root test
- 5.4. Absolute convergence
- 5.5. Power series
- 5.6. Radius and interval of convergence

6. Differential Equations

(7 hours)

- 6.1. Power series method
- 6.2. Legendre's equation
- 6.3. Legendre polynomial function; Properties and applications
- 6.4. Bessel's equation and Bessel's function

Tutorial

(30 hours)

- 1. Techniques of partial differentiation, differentiation of composite and implicit functions, total derivatives, and related exercises
- 2. Exercises related to Euler's theorem
- 3. Exercises related to extreme values of two and three variables
- 4. Change of order of integration in multiple integrals
- 5. Exercises related to application of double and triple integrals in finding area, volume, moment of inertia, mass and centroid
- 6. Exercises related to Analytical Geometry of three dimensional
- 7. Exercises on line integrals, independent of path
- 8. Exercises on surface integrals
- 9. Exercises on Green's theorem, verification and application in calculating line integrals
- 10. Verification of Gauss' Divergence theorem, Stoke's theorem, and application in calculating surface integrals
- 11. Test the example of convergence and divergence of the sequence and find the interval and radius of convergence
- 12. Exercises related to Bessel's function and Legendre's Polynomial

Reference books:

- 1. E. Kreyszig, "Advance Engineering Mathematics", Wiley, New York.
- 2. G.B. Thomas, and R.L. Finney, "Calculus and Analytic geometry", Addison Wesley.
- 3. G.D. Pant, G. S. Shrestha, "Integral Calculus and Differential Equations", Sunila Prakashan, Nepal
- 4. M. B. Singh, B. C. Bajrachrya, "Differential calculus", Sukunda Pustak Bhandar, Nepal
- 5. M. B. Singh, B. C. Bajrachrya, "A text book of Vectors", Sukunda Pustak Bhandar, Nepal
- 6. S. M. Maskey, "Calculus", Ratna Pustak Bhandar, Nepal
- 7. Y. R. Sthapit, B. C. Bajrachrya, "A textbook of Three-Dimensional Geometry", Sukunda Pustak Bhandar, Nepal.

Evaluation Scheme

The questions will cover all the chapters of the syllabus. The evaluation scheme will be as indicated in the table below:

Chapter	Hours	Marks Distribution*
1.	4	6
2.	6	8
3.	11	14
4.	12	16
5.	5	6
6.	7	10
Total	45	60

* There may be minor deviation in marks distribution.